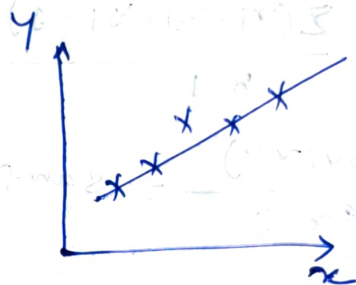
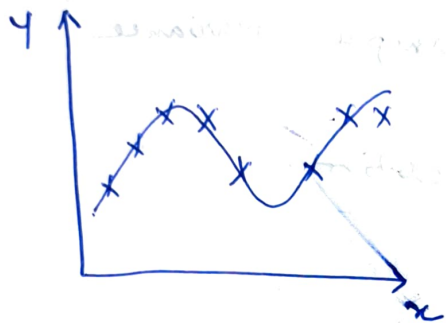


Regression

Correlation tells us if there is an association between x and y but it does not allow to predict one variable from the other.

However, regression allow us.

Start with the problem of polynomial curve fitting:



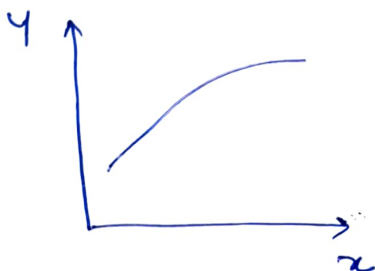
$x \rightarrow$ Independent variable

$y \rightarrow$ dependent variable

Scenario 1

If you know the equation

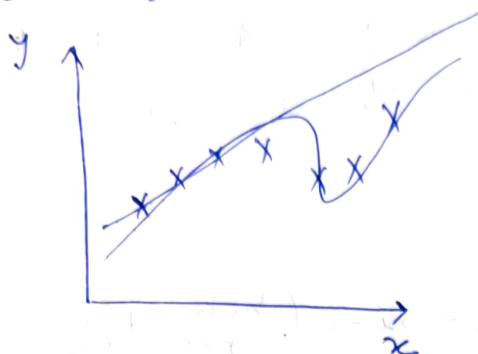
$$y = 2 + 3x + 4x^2$$



Can draw the curve

Scenario 2

- points are given.
- polynomial structure is not known
- we have to approximate the polynomial structure from some points.



So many structures/
equation

Some points $\rightarrow$ training samples $\langle x_i, y_i \rangle$

x_i	y_i
0.1	5
0.6	2
0.9	1.2

Find equation (best fit one)

Example of regression problem (house price prediction)

Size of house	price
600	15
1120	30
⋮	40
⋮	⋮

Back to polynomial curve fitting problem

$$y = a_0 + a_1 x^1 + a_2 x^2 + \dots + a_n x^n$$

or

$$y = w_0 + w_1 x^1 + w_2 x^2 + \dots + w_n x^n$$

$\rightarrow$ we want to find this equation from training sample $\langle x_i, y_i \rangle$

primarily unknown

$n \rightarrow$ degree of polynomial

$w_0, w_1, \dots, w_n \rightarrow$ parameters

Supervised Learning

- There is a set of training samples $\langle x_i, y_i \rangle$
- The goal is to learn the function/hypothesis $h(x_i)$ so that it is a good predictor for the corresponding value of y using training samples.

$$h: x \rightarrow y$$

$y \rightarrow$ Continuous / Regression (house price prediction)

$y \rightarrow$ discrete / Classification (spam/no-spam identification)

Aim: to design learning algorithm, to approximate hypothesis structure

Linear Regression

Assume: y is a linear function of x

$$\hat{y} = h(x) = w_0 + w_1 x$$

Learn $h(x)$ from $\langle x_i, y_i \rangle$ training sample

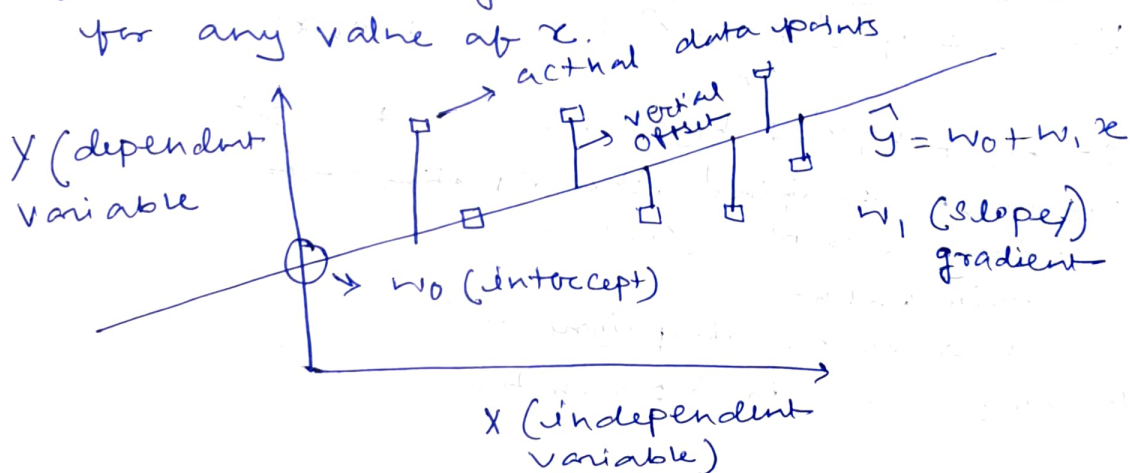
m = number of training sample

n = number of feature = 1

Aim: fit a straight line,

$$\hat{y} = h(x) = w_0 + w_1 x$$

to data that gives best prediction of y for any value of x .



Find best fit line that minimize difference/distance between data and fitted line i.e the residual (ϵ)

$$\epsilon_i = y_i - \hat{y}_i$$

$\downarrow$ $\downarrow$
 actual predicted

$i = 1, 2, \dots, m$

$\downarrow$
 number
 of
 training samples

- There are so many such lines or so many assumption about equation

$$\hat{y} = 2 + 3x \quad | \quad \hat{y} = 10x$$

- Find the line having minimum residual.
- For that, we have to tune w_0, w_1

Ordinary least square regression (OLS)

- To find the best fit line that minimize the sum of the square of residuals.

- hypothesis : $\hat{y} = h(x) = w_0 + w_1 x$

- Residual : $\epsilon = y - \hat{y}$

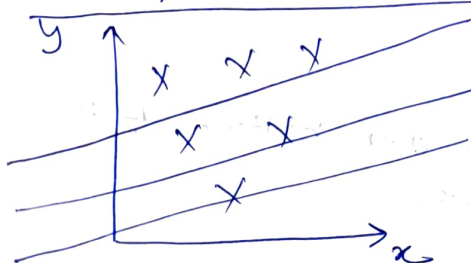
- The sum of square of residual = $\sum (y - \hat{y})^2$

- Find w_0, w_1 that minimize

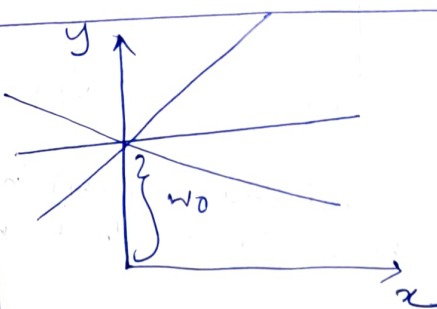
$$LSE = \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

Now, we are searching for w_0, w_1

Finding w_0 (changing y intercept value)



Finding w_1 (changing slope)



Simple linear regression

One independent variable (x)

One dependent variable (y)

Model: $\hat{y} = h(x) = w_0 + w_1 x$

Residual: $(y - \hat{y})$

We have

training samples

$$\langle x_i, y_i \rangle$$

$$i = 1, 2, \dots, m$$

Minimize

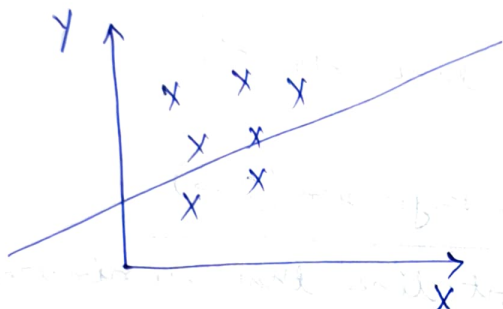
$$J(w_0, w_1) = \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

RSS: Residuals square

LSE: Least square error

$$J(w): w = [w_0 \ w_1]$$

Find the best fitted line



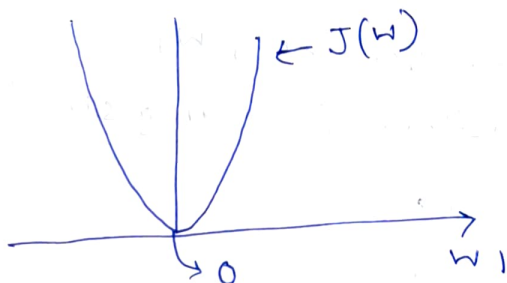
$$\hat{y} = w_0 + w_1 x$$

↓ ↓
y intercept slope

- Sometime, we use $\frac{1}{2}$ before LSS for convenience and simplicity. It does not hamper the minimization.

$$J(w) = \frac{1}{2} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

- If we plot $J(w)$ with respect to a parameter (slope w_1) yield the classic U shaped parabola because it is a squared term.



- So, the minimum $J(w)$ is at the bottom of the curve, where the gradient is zero.

- We can find ~~min~~ w_0, w_1 that give min $J(w_0, w_1)$ by taking partial derivative of $J(w)$ with respect to w_0, w_1 separately.

Approach 1 Ordinary linear regression (OLS)

A closed form solution in terms of linear regression, refers to an analytical method to find the exact coefficient of polynomial by minimizing the error without iterative optimization.

$$\frac{\partial}{\partial w_0} J(w) = 0$$

$$\frac{\partial}{\partial w_1} J(w) = 0$$

$$\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x} = \frac{\sum_{i=1}^m y_i - \sum_{i=1}^m \hat{w}_1 x_i}{m}$$

$$\hat{w}_1 = \frac{\text{cov}(x, y)}{s_x^2} = r_{xy} \frac{s_y}{s_x}$$

• There is a relation between slope and correlation coefficient.

• There is a relation between slope and s_x and s_y .

Multiple linear regression

One dependent variable (y)
Multiple independent variable ($x_1, x_2, \dots, x_n$)